CSC 212—Algorithms and Complexity
Run some experiments using the RSA algorithm.
Practical 4 Term 4 3 October 2016

Due by 17h30 on 9 October 2016

There are several separate but very simple tasks. (1) Familiarize yourself with the RSA algorithm by rewriting the code given to you in `python` to java and run it for $p = 3$ and $q = 11$. First use the value of $e = 7$ and find d using the supplied function for the inverse if $e \bmod]phi(n)$ to calulate d from e and $\phi(n)$ using $d = \text{inverse}(e, \phi(n))$. Try to calculate all the values of e such that $1 < e < \phi(n)$ and each corresponding value of d. Do the messages change for different (e, d) pairs? (2) Once you are familiar with RSA try to use any different pair of primes p and q to calculate $n = p \times q$, and $phi(n) = (p - 1) \times (q - 1)$, then for every possible qualifying e, such that $1 < e < \phi(n)$, get all the qualifying d values using $d = \text{inverse}(e, \phi(n))$ and inspect the codes that are generated and draw some conclusions.

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1. Note that this practical does not have the usual format of coding something quickly and accurately. We want you to study the numbers that are produced and write up your observations. This is a simple coding excercise in which you are required to manipulate the numbers that the RSA algorithm produces and then decide for yourself what is going on.

2. Handing in both code working code for this practical and giving an understandable write up of at least 300 words about your results is required. Make a subdirectory for your write up called `writeup`. Put your writeup between the lines `\begin{document}` and `\end{document}` in the a directory supplied in the `../notes/ds` file called `54writeup`. Simply copy the whole directory into your `54practical` directory. Also replace the words `myname` and my `studentnumber` with your *first name* followed by your *surname,* and your *student number.* To see the output this produces you may run `pdflatex 54writeup` or simply run `make` in the correct directory. The output is a `.pdf` file called *54writeup.pdf.*

3. `cd; make submit` process. The `Makefile`s are in `/export/home/notes/ds/makefiles/`. The one goes into your home directory and the other must be in your current practical's directory, i.e., `54practical`.
